

Project Action Plan & Scope Realignment

Addressing Proposal Defense Evaluation Feedback & Engineering Roadmapping

Doc ID: CS-IS_[group_number].pdf
Date: September 2026
Supervisor: Mr. Tharindu Ranathunga

1 TARGET BUSINESS DOMAIN	4 UNDERGRADUATE DEVS	3 LANGUAGES (SINHALA/TAMIL/EN)	32 Wks DEVELOPMENT RUNWAY
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Executive Summary: During the initial proposal defense evaluation, the panel raised critical concerns regarding project feasibility, broadness of scope, and unclear target user profiles in our proposed generic multi-domain "Smart ERP." In accordance with the panel's guidance, this Action Plan establishes an immediate pivot from an unmanageable general-purpose system to a highly focused, localized **Point-of-Sale & Inventory Optimization System tailored exclusively for Independent Sri Lankan Retail Groceries.**

1. PANEL FEEDBACK SYNTHESIS & EVALUATION ANALYSIS

Feedback Category	Specific Panel Critique (Lead & Second Panelists)	Risk / Impact Identified
1. System Scope & Feasibility	Attempting 10+ disparate ERP modules across 4 unrelated domains (pharmacies, groceries, hardware, retail) in a 4-person student team within one year is an unrealistic, "Sisyphean task."	Critical Severity Surface-level modules with negligible depth; guaranteed failure during code defense.
2. "Reinventing the Wheel"	Mature, feature-rich open-source ERPs (e.g., ERPNext, Odoo) already exist for SMEs. Building a generic ERP from scratch lacks novelty and academic rigor.	High Risk Panel inability to verify originality; suspicion of wrapping existing open-source codebases.
3. Domain Definition & Client Absence	"Small/Medium Enterprises" was vague without concrete constraints (headcount, inventory SKU volume, catalog depth, workflow specifics).	Scope Ambiguity Inability to formulate clear verification metrics or meaningful user stories.
4. Module Gaps (HR & POS)	If an ERP includes "HR," it cannot merely be an attendance tracker—it requires payroll, appraisals, and onboarding. Otherwise, it is misclassified.	Architectural Mismatch Superficial feature implementations that fail standard ERP classification criteria.
5. Market & Tech Assumptions	Assuming Sri Lankan SMEs do not use payment gateways is factually incorrect. Localization requires true multilingual workflows, not simple strings.	Implementation Gap Incomplete localization logic and misaligned business requirements.

2. SUPERVISORY CONSULTATION & ALIGNMENT

Following the presentation, the team consulted Project Supervisor **Mr. Tharindu Ranathunga** on September 23, 2026. The supervisor endorsed the panel's recommendation to abandon the multi-domain generic architecture. It was mutually agreed that narrowing down to a single domain provides the depth required for a strong Second Year Group Project while guaranteeing academic integrity and delivery feasibility.

3. COMPREHENSIVE ACTION TAKEN: THE STRATEGIC PIVOT

We have completely restructured the proposal scope from a generic multi-tenant ERP platform into a purpose-built domain solution: **SmartGrocer LK — Retail Intelligence & Automated Inventory System for Neighborhood Groceries**. The core comparative matrix below illustrates our structural pivot:

Original Proposal (Rejected Scope) De-scoped • Architecture: Generic multi-domain ERP attempting to service 4 sectors (Pharmacies, Groceries, Hardware, Retail). • Modules: 10+ modules: Finance, General HR, SCM, Warehouse, CRM, Multi-warehouse, Procurements. • Domain Boundary: Ill-defined SME classification with no ceiling on employees, branches, or catalog size. • Core Novelty: Generic configurability with vague "smart" analytical branding.	Revised Action Plan (Approved Focus) Active Blueprint • Architecture: Single-domain, modular retail management tailored specifically for FMCG independent grocers. • Modules: 4 deeply engineered modules: POS Checkout, Smart Batch Inventory, Supplier Reordering, Cash Flow. • Domain Boundary: 1–3 terminal grocery outlets, ≤10 staff, 2,500–5,000 active SKUs, fast-turnover FMCG. • Core Novelty: Native tri-lingual billing (Sinhala/Tamil/English), auto-stock forecasting, LankaQR integration.
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4. SYSTEMATIC REMEDIATION PLAN BY EVALUATION POINT

Feedback Item	Action Taken / Design Modification	Verification & Deliverable Evidence
1. Single Domain Selection	Selected Independent Grocery Outlets . Groceries utilize standardized EAN/UPC barcodes, predictable supplier cycles, and high transaction volume.	Formal Requirements Specification document mapped to real-world workflows of a partner retail store in suburban Colombo.
2. Scope & Module Pruning	Completely eliminated complex Enterprise HR (recruitment, benefits), multi-tier warehouse routing, and manufacturing logic from the codebase.	Simplified domain schema reducing total database tables from 48 down to 18 tightly integrated, highly normalized relational entities.
3. Justification for Custom Build	Rather than reproducing standard ERP tables, focused on local grocery friction: Sinhala/Tamil POS receipts, weighted-scale inputs, and expiry alerts.	Benchmarking matrix contrasting SmartGrocer LK against standard ERPNext to prove local efficiency, lightweight deployment, and speed.
4. Sri Lankan Localization	Integrated structured UTF-8 JSON language packs (Sinhala, Tamil, English) for UI & receipt generation; added dynamic LankaQR digital payment flow.	Fully functional bilingual thermal receipt preview generator and LankaQR payload validation test suites.
5. Redefining "Smart" Features	Replaced broad claims with an explicit, lightweight Safety Stock & Expiry-Risk Heuristic Algorithm for perishable groceries.	Documented algorithmic formula calculating Reorder Point (\$ROP\$) and Wastage Prevention alerts based on moving average daily sales.

5. REVISED SYSTEM BOUNDARY & ENTITY SIZING

To eliminate ambiguity regarding "business size" highlighted by Examiner 2, the system's operational envelope is bounded as follows: **(a) Hardware Target:** Low-cost desktop/laptop running a local PHP/MySQL or SQLite stack with USB barcode scanner, digital scale, and 80mm ESC/POS thermal printer. **(b) Store Capacity:** Stores handling 150–600 daily retail transactions, up to 10 staff members (cashiers, shift leads, owner-admin), and up to 5,000 inventory items.

6. TARGET SYSTEM ARCHITECTURE & MODULE ALLOCATIONS

Decomposition of work across four undergraduate team members to ensure individual accountability.

Module 1: Barcode POS & Localized Billing Student 1 (Lead) High-speed keyboard/scanner-driven checkout terminal designed for rapid FMCG retail transactions. - USB barcode parsing with instant lookup (<100ms response). - Dual-currency/payment handling (Cash, LankaQR digital rails). - Bilingual ESC/POS receipt generation (Sinhala & Tamil UTF-8 fonts). - End-of-day register reconciliation (Z-Reports & cash float balance).	Module 2: Smart Inventory & Expiry Tracking Student 2 Stock ledger emphasizing batch numbers, wholesale packs, and perishable item lifecycles. - First-Expired-First-Out (FEFO) automated batch prioritization. - Expiry countdown alerts (30-day, 14-day, 48-hour color coding). - Unit conversions (e.g., purchasing 25kg bulk bags, selling in 500g). - Inventory write-off & shrinkage logging with supervisor sign-off.
Module 3: Demand Forecast & Reorder Engine Student 3 Algorithmic calculation of purchase orders based on rolling sales velocities to prevent stockouts. - Dynamic Reorder Point (\$ROP\$) based on supplier lead times. - Automated Purchase Order (PO) draft generation per local distributor. - Distributor delivery verification against purchase order line items. - Fast-moving vs. slow-moving stock categorizer (ABC Analysis).	Module 4: Cash Flow, Credit & Shift Ops Student 4 Simplified financial ledger handling grocery trade credit and shift-based cashier tracking. - Customer credit book ("Potha" system) ledger with balance limits. - Cashier shift logins, cash drawer audits, and discrepancy logs. - Supplier payables tracking with settlement due date reminders. - Daily gross profit and category-wise margin reporting.

7. CONCRETE ALGORITHMIC JUSTIFICATION ("SMART" COMPONENT)

Addressing Examiner 1’s critique on our previous misuse of the term "Smart ERP," we have defined an exact, computable inventory replenishment heuristic. Rather than claiming generic AI, Module 3 executes a **Dynamic Buffer Reorder Point (DB-ROP)** calculation computed per inventory SKU:

Reorder Point Formula & Safety Stock Buffer: $ROP = (D_{avg} \times L) + (Z \times \sigma_L \times \sqrt{L})$ Where: D_{avg} = Average Daily Sales over 14 days, L = Supplier Lead Time in days, Z = Service Level Factor (1.65 for 95% availability), and σ_L = Standard deviation of daily sales. This mathematical rigor replaces vague smart marketing claims with functional software engineering.

8. SOFTWARE ARCHITECTURE & TECHNOLOGY STACK

Layer	Technology Selected	Architectural Justification
Backend API	PHP 8.2 (Laravel Framework)	Strict MVC structure, robust Eloquent ORM, built-in validation, clean RESTful endpoints.
Frontend UI	React.js with Tailwind CSS	SPA architecture for lightning-fast cashier terminal interactions without page reloads.
Persistence	MySQL 8.0 (InnoDB Engine)	ACID transactional guarantees essential for inventory count decrements and cash balances.
Localization	JSON I18n Data Tables	Dynamic frontend locale switching; verified Unicode UTF-8 output for thermal receipt printers.

9. WORK BREAKDOWN & MILESTONES (ROADMAP TO INTERIM & FINAL)

To ensure strict feasibility within the 32-week academic schedule, the project is scheduled across four synchronized development sprints leading directly to the Interim Evaluation and Final Defense:

Timeline	Milestone Phase	Key Engineering Deliverables	Status / Verification
Weeks 1–4 Oct 2026	Phase 1: Domain Spec & Architecture	Gather on-site workflows from 2 grocery partners; draft ER diagrams, API specs, and finalize revised proposal document.	Completed Revised Proposal Approved
Weeks 5–12 Nov–Dec 2026	Phase 2: Core POS & Inventory Base	Implement barcode scanner pipeline, POS cart state, stock master schema, and Sinhala/Tamil thermal printing driver.	In Progress Alpha Build Ready
Weeks 13–18 Jan–Feb 2027	Phase 3: Interim Defense Milestone	Deliver complete end-to-end checkout loop, batch expiry tracking, and preliminary LankaQR integration. Interim evaluation.	Milestone 50% Working Prototype
Weeks 19–26 Mar–Apr 2027	Phase 4: Smart ROP & Customer Credit	Incorporate Reorder Point calculation engine, Customer Credit book ("Potha"), supplier PO generator, and cash audit trails.	Milestone Feature Freeze
Weeks 27–32 May–Jun 2027	Phase 5: Field Testing & Final Defense	Deploy pilot hardware at partner store; run stress testing (1,000 continuous checkouts); assemble technical documentation.	Final Target 100% Production Ready

10. RISK MANAGEMENT & FALLBACK STRATEGY

Identified Risk	Potential Failure Mode	Proactive Mitigation / Fallback Measure
Thermal Printer Drivers	ESC/POS rendering errors when printing complex Sinhala/Tamil glyphs.	Pre-render receipt lines into monochromatic raster bitmaps on the server before transmitting RAW commands to the printer.
Scope Creep During Development	Client store requesting warehouse logistics or payroll features during pilot.	Enforce strict requirement freeze per this Action Plan; all secondary requests logged under "Future Work."
Academic Load Imbalance	Concurrent coursework deadlines impeding group development velocity.	Bi-weekly Git milestone reviews; decoupled modules allowing parallel, unblocked development by all 4 members.

11. STUDENT GROUP COMMITMENT & SIGN-OFF

The group hereby formally submits this Action Plan to the project evaluation committee and supervisor. We confirm that all feedback items delivered during the proposal presentation have been evaluated, acknowledged, and addressed through this focused re-scoping.

[Student 1 Name] Index: [Index 1] Module 1 (POS Terminal)	[Student 2 Name] Index: [Index 2] Module 2 (Inventory)	[Student 3 Name] Index: [Index 3] Module 3 (Forecasting)	[Student 4 Name] Index: [Index 4] Module 4 (Credit & Cash)
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